

Research Methods

Day 8 · Understanding Data & Beginning Analysis

Outline for Today

1. **Introduction Presentations: "The Beginning"**
2. **What Is Data? Types & Sources**
3. **Data Quality & Common Mistakes**
4. **Accuracy in Data Recording**
5. **Data Warm-Up & Your Data Story**

Welcome to Day 8!

Today you meet **your data** — and start turning it into findings.

By the end of the day you will be able to:

- Build **intuition about patterns** in your team's dataset
- Present the **introduction** of your project with stage comfort
- Recognize **types of data, common errors**, and recording best practices
- Begin connecting **data findings to your research question**





Introduction Presentations: "The Beginning"

Introduction Presentations 40 min

Each team presents its **introduction slides** (from the Session 7 homework).

Structured feedback format:

- 1 **2 minutes** to present.
- 2 **2 minutes** of peer feedback using the **Presentation Feedback Checklist** (Session 6 tool).
- 3 The instructor adds **one specific observation** per team.

 Feedback is **specific and kind** — name one thing that worked and one thing to sharpen.

40:00



What Is Data?

Types & Sources

Focus of Day 8

 **Understanding** different types of data.

 **Avoiding** biases in data collection.

 **Recording** data: best practices.

 **Reporting** data: tools to use.



What Is Data?



Evidence

Information collected to examine hypotheses, address questions, or describe phenomena.



Foundation

Transforms uncertainty into insight grounded in observable reality.



Replicable Conclusions

Moves beyond opinion toward supported, replicable conclusions.

Quantitative vs. Qualitative Data

Quantitative

Information that can be **measured or counted** — expressed in numbers.

- Test scores
- Survey ratings
- Hours spent on homework
- Growth measurements

Qualitative

Descriptive, non-numerical information — provides context and depth.

- Open-ended survey responses
- Observational notes
- Interview narratives
- Written explanations

A Balanced Approach



Quantitative Precision

Measures patterns with numerical accuracy.



Qualitative Depth

Explains the "*why*" behind the patterns.



Integrated Research

Combines approaches for a comprehensive understanding.

A Balanced Approach



Quantitative Precision

Measures patterns with numerical accuracy.



Qualitative Depth

Explains the “*why*” behind the patterns.



Integrated Research

Combines approaches for a comprehensive understanding.

The best projects mix both — numbers show *what* is happening; words and context explain *why*.

Primary vs. Secondary Data

Primary Data

Information you collect **directly**, through your own effort.

- Experiments
- Surveys
- Interviews
- Observations

Secondary Data

Information that **already exists**, collected by others.

- Books & articles
- Websites
- Databases
- Government / agency records

When to Use Each Type

1 Use Primary When...

Your question is specific and unique · you need fresh data nobody has · you want control over how it's collected.

2 Use Secondary When...

The information already exists · you need to save time · you need broader context.

3 Best Practice

Many strong projects use **both** types for a fuller understanding.

Data Structure & Source

Structure — how it spans time:

- **Cross-sectional:** many units at **one** point in time (a snapshot).
- **Longitudinal:** the **same** units tracked **over time** (a movie).

Source — where it comes from:

- **Administrative:** records kept for operations (school enrollment, tax records).
- **Survey:** collected by asking people questions.
- **Experimental:** generated under controlled, randomized conditions.

Knowing the structure and source tells you what claims the data can — and cannot — support.

Data Quality Issues to Watch For

 **Missing data** — *why* it's missing matters:

- **Missing at random** — gaps unrelated to the value itself; usually less harmful.
- **Missing not at random** — the missingness is *related* to the value (e.g., high earners skip the income question). This **biases** your results.

 **Measurement error** — the recorded value differs from the true value (faulty instrument, vague question, typos).

 **Historical limitations** — old data may use different definitions, categories, or methods than today — compare with care.

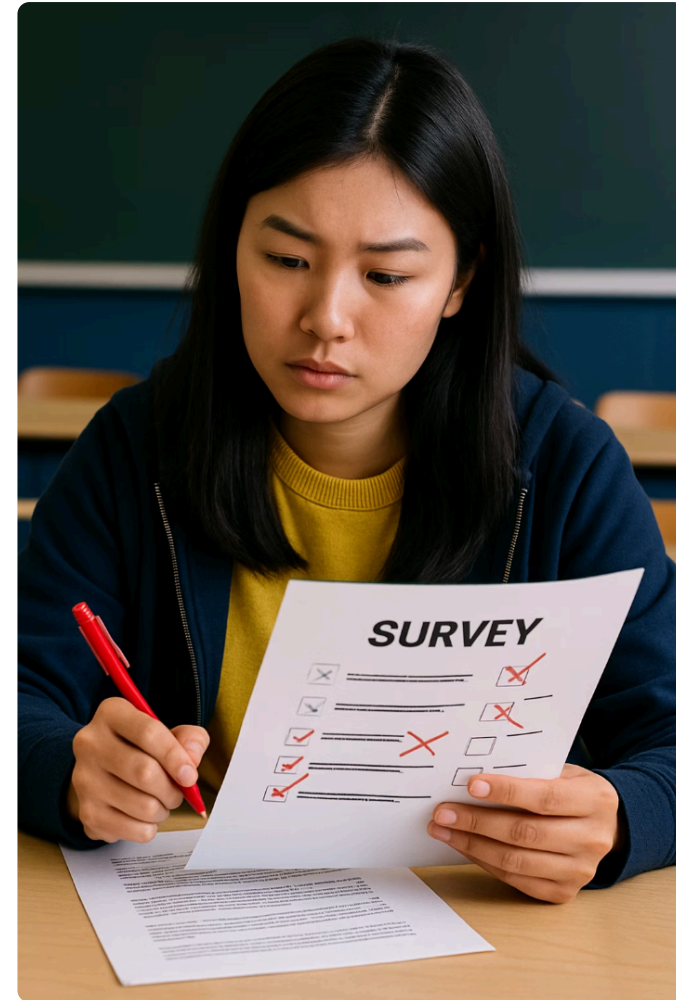


Data Quality & Common Mistakes

Common Mistakes in Research

We can make mistakes at **every stage**:

- **Selecting participants** → *selection bias*
- **Collecting data** → *measurement & response bias*
- **Analyzing data** → *confirmation bias*
- **Reporting results** → *reporting bias*



Biased Sampling & Why Random Sampling Helps

✗ Biased sampling

Example: only surveying students who play sports about their favorite activities.

Fix: include students with diverse interests and backgrounds — use **random sampling** where possible.

✂ Why random sampling helps

- **Representative results** — findings reflect the whole population.
- **Reduced bias** — everyone has an equal chance of selection.
- **Statistical validity** — supports more accurate analysis.

Fixing Biased (Leading) Questions

A **leading question** pushes the respondent toward an answer. Rewrite it to be **neutral**.

✗ Biased question	✓ Neutral alternative
"Don't you agree that school starts too early?"	"What time do you think school should start?"
"Why is the new lunch policy so unfair?"	"What are your thoughts on the new lunch policy?"
"How much do you hate doing homework?"	"How would you describe your experience with homework?"
"Wouldn't you prefer more recess time?"	"What is your opinion about the current recess time?"

Confirmation Bias

The trap

Only noticing the data that **supports** your hypothesis — and overlooking the rest.

The fix

Actively look for evidence that **contradicts** your expectations. Let the data change your mind.

*A good researcher tries to prove themselves **wrong** — not just right.*



Accuracy in Data Recording

Why Accurate Recording Matters

✓ **Valid conclusions** — accurate data leads to trustworthy findings.

📄 **Replicable research** — others can verify and build on your work.

⚖️ **Research integrity** — honest recording reflects scientific ethics.



Data Recording Best Practices

Record Immediately

Don't rely on memory — write down observations right away.

Use Standardized Forms

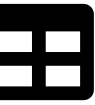
Create consistent templates for data collection.

Document Visually

Take photos or videos when appropriate.

Back Up Your Data

Keep multiple copies in different locations.



Data Warm-Up & Your Data Story

Session 8 Mission Sheet: Data Warm-Up 30 min

As a team, **explore your dataset** using the Session 8 Mission Sheet.

- 1 **Count** the rows and columns.
- 2 **Describe** your key variables.
- 3 Note **one surprising observation**.
- 4 Note any **missing or unusual values**.
- 5 Calculate the **mean and range** of one numerical variable.

30:00

Scientific Storytelling: Data Story Opening Lines 15 min

Write the **opening line** of your Results section. It must be **specific and factual**:

*"Our dataset contains **[N]** observations from **[source, year]**. The average **[DV]** in our sample is **[value]**."*

Then **read it aloud** — practice precision.

🎯 **Precise, not vague.** "About 500-ish students" → "Our dataset contains **512** students surveyed in **2023**."

15:00



Homework & After-Class Work

After-Class Team Session ⚙️ 3:15 – 5:15 PM

Teams work with a **TA / peer mentor** available. Produce a **tangible output before dinner** – *not free time*.

1

Data Warm-Up Mission Sheet: complete all 4 sections with your actual dataset – variable table, initial observations, descriptive stats, curiosity questions.

2

Build a **descriptive statistics summary table** (Python/Excel): mean, median, SD, range for all key numerical variables.

3

Identify and document any **data quality issues** (missing values, outliers, format errors) in your **data cleaning log**.

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🚩 **Team output due before dinner:** descriptive statistics table + data cleaning log + completed Session 8 Mission Sheet.

Evening Work Period 📅 7:00 – 9:00 PM

Individual + partner work during CTD RTA study hours. TA available for drop-in questions.

- **Individual DataCORE Journal:** describe the **shape** of your data distribution. Is it skewed? Are there outliers? What might explain them?
- **(RcS):** create a **correlation matrix** for your key variables. Screenshot and annotate — which correlations are strongest? Do they make sense theoretically?
- **Slide update:** revise your **Methods slide** with what you now know about your data — be specific about dataset name, timeframe, and sample.

